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1  /* =====
2  2.WITHDRAW MONEY
3  ATM transactions are additionally recorded in ATMTransactions
4  =====*/
5  CREATE OR ALTER PROCEDURE Withdraw_Money
6  (
7      @AccountId          INT,
8      @WithdrawAmount     DECIMAL(18,2),
9      @Category           VARCHAR(30),    --( Cash Withdrawal | ATM Withdrawal
                                         | Cheque Clearance | Charges)
10     @Channel            VARCHAR(20),    --( Branch | ATM | Internet Banking |
                                         Mobile Banking | POS | Auto Debit)
11     @ATMID              INT = NULL
12 )
13 AS
14 BEGIN
15     SET NOCOUNT ON;
16
17     BEGIN TRY
18         BEGIN TRANSACTION;
19
20         -- 1. Validate withdrawal amount
21         IF @WithdrawAmount <= 0
22         BEGIN
23             THROW 50001, 'Withdrawal amount must be greater than zero.', 1;
24         END;
25
26         -- 2. Validate withdrawal category
27         IF @Category NOT IN
28         (
29             'Cash Withdrawal',
30             'ATM Withdrawal',
31             'Cheque Clearance',
32             'Charges'
33         )
34         BEGIN
35             THROW 50003, 'Invalid withdrawal category.', 1;
36         END;
37
38         -- 3. Validate account
39         IF NOT EXISTS
40         (
41             SELECT 1
42             FROM Accounts
43             WHERE AccountID = @AccountId
44                   AND AccountStatus = 'Active'
45         )
46         BEGIN
47             THROW 50002, 'Account does not exist or is not active.', 1;
48         END;
49
50         -- 4. Validate Category + Payment Channel
51         IF (@Category = 'Cash Withdrawal' AND @Channel NOT IN ('Branch', 'ATM'))
52         OR
53         (@Category = 'ATM Withdrawal' AND @Channel <> 'ATM')
54         OR
55         (@Category = 'Cheque Clearance' AND @Channel <> 'Branch')
56         OR
57         (@Category = 'Charges' AND @Channel NOT IN ('Branch', 'ATM', 'Internet Banking',
58             'Mobile Banking', 'POS', 'Auto Debit'))
59         BEGIN
60             THROW 50004, 'Invalid Category and Payment Channel combination.', 1;
61         END;
62
63         -- 5. Validate ATM details when transaction is through ATM
64         IF @Channel = 'ATM'
65         BEGIN
66             IF @ATMID IS NULL
67             BEGIN
68                 THROW 50005, 'ATMID is required for ATM transactions.', 1;
69             END
70
71             IF NOT EXISTS
72             (
73                 SELECT 1
74                 FROM ATM
75                 WHERE ATMID = @ATMID
76                       AND Status = 'Active'
77             )
78             BEGIN
79                 THROW 50006, 'ATM does not exist or is not active.', 1;
80             END
81         END;
82
83         -- 6. ATMID must be NULL for non-ATM transactions
84         IF @Channel <> 'ATM' AND @ATMID IS NOT NULL
85         BEGIN
86             THROW 50007, 'ATMID must be NULL for non-ATM transactions.', 1;
87         END;
88
89         -- 7. Get account balances
90         DECLARE @OpeningBalance DECIMAL(18,2);
91         DECLARE @AvailableBalance DECIMAL(18,2);
92         DECLARE @OverdraftLimit DECIMAL(18,2);
93
94         SELECT
95             @OpeningBalance = CurrentBalance,
96             @AvailableBalance = AvailableBalance,
97             @OverdraftLimit = OverdraftLimit
98         FROM Accounts
99         WHERE AccountID = @AccountId;
100
101         -- 8. Validate sufficient funds
102         IF (@AvailableBalance + @OverdraftLimit) < @WithdrawAmount
103         BEGIN
104             THROW 50008, 'Insufficient funds for this withdrawal.', 1;
105         END;
106
107         -- 9. ATM cash validatces
108         DECLARE @ATMOpeningBalance DECIMAL(18,2);
109         DECLARE @ATMClosingBalance DECIMAL(18,2);
110
111         IF @Channel = 'ATM'
112         BEGIN
113             SELECT
114                 @ATMOpeningBalance = CashBalance
115             FROM ATM
116             WHERE ATMID = @ATMID;
117
118             IF @ATMOpeningBalance < @WithdrawAmount
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